

Comparative Sequence Analysis of Independently Evolved Luciferases

Project Proposal

Instructor:

Ms. Rushda Muneer

Group Members:

Eman Naeem	23F-0808
Afnan Asif	23L-0709
Saad Riaz Taqi	23L-0669

National University Of Computer and Emerging Sciences
Department of Computer Science
Lahore, Pakistan

1. Abstract

Bioluminescence is the ability of living organisms to produce light. It occurs in organisms such as fireflies, marine animals, bacteria, and fungi. Interestingly, these organisms do not all use the same light-producing protein. Many luciferase proteins evolved independently from different protein families while performing a similar function, making bioluminescence an example of convergent evolution.

This project will computationally compare independently evolved luciferase proteins from fireflies, *Renilla*, luminous bacteria, and fungi. Protein sequences will be collected from public databases such as UniProt and NCBI in FASTA format. Sequence alignment will be used to measure similarity between luciferase families. Protein properties including length, molecular weight, amino acid composition, hydrophobicity, aromaticity, and isoelectric point will also be calculated.

Where suitable data are available, selected luciferases will be compared with related non-luminous proteins. Results will be presented using similarity matrices, heatmaps, graphs, and comparative tables. The objective is to determine whether independently evolved luciferases share common molecular characteristics despite having different evolutionary origins.

2. Introduction

Bioluminescence is a biochemical process in which organisms produce visible light. In many systems, proteins called luciferases help catalyze light-producing reactions. Bioluminescence has evolved independently in insects, marine organisms, bacteria, and fungi. Their luciferases can belong to different protein families despite performing a similar function. The main research question is, do independently evolved luciferases share common sequence or physicochemical characteristics despite having different evolutionary origins?

3. Literature Review

Previous studies show that luciferases do not belong to one universal protein family. Delroisse et al. (2021) compared luciferases and photoproteins from many organisms and identified several evolutionary groups. He et al. (2024) showed that firefly luciferases are related to the acyl-CoA synthetase family. Research on brittle stars has shown that luciferase activity can also evolve from haloalkane dehalogenase-related proteins. Fungal bioluminescence uses another distinct protein called Luz. These studies show that different protein families can independently evolve toward the same general function of light production.

4. Proposed Methodology / Tentative Algorithms and Tools

Protein sequences will be collected from UniProt and NCBI in FASTA format. The dataset will include representative luciferases from:

- Fireflies
- *Renilla*
- Luminous bacteria (LuxA)

- Bioluminescent fungi (Luz)

Pairwise protein sequence alignment and similarity analysis will be performed using Needleman-Wunsch, Smith-Waterman, and BLASTP through Biopython and established bioinformatics tools.

For each protein, the following properties will be calculated:

- Protein length
- Molecular weight
- Amino acid composition
- Hydrophobicity
- Aromaticity
- Isoelectric point

Selected luciferases will be compared with related non-luminous proteins. Protein-family and domain information will be checked using InterPro.

Tools: Python, Biopython, Pandas, NumPy, Matplotlib, SciPy, and InterPro.

5. Expected Outcome or Deliverables

The project will produce:

- A curated FASTA dataset
- Python scripts for sequence and protein analysis
- Pairwise alignment results
- Similarity matrices and heatmaps
- Physicochemical comparison graphs
- Comparison with selected non-luminous proteins
- Final conclusions based on observed patterns

The analysis will determine whether independently evolved luciferases remain distinct at the sequence level while showing any common physicochemical patterns despite their different evolutionary origins and shared light-producing function.

6. References

[1] J. Delroisse, L. Duchatelet, P. Flammang, and J. Mallefet, “Leaving the Dark Side? Insights Into the Evolution of Luciferases,” *Frontiers in Marine Science*, vol. 8, Art. no. 673620, 2021. [Online]. Available: https://doi.org/10.3389/fmars.2021.673620 Frontiers

[2] J. He *et al*., “Multiple Origins of Bioluminescence in Beetles and Evolution of Luciferase Function,” *Molecular Biology and Evolution*, vol. 41, no. 1, Art. no. msad287, 2024. [Online]. Available: https://doi.org/10.1093/molbev/msad287 PubMed Central (PMC)

[3] A. A. Kotlobay *et al*., "Genetically Encodable Bioluminescent System from Fungi," *Proceedings of the National Academy of Sciences*, vol. 115, no. 50, pp. 12728–12732, 2018. [Online]. Available: https://doi.org/10.1073/pnas.1803615115 PubMed Central (PMC)

[4] E. S. Lau *et al*., "Functional Characterization of Luciferase in a Brittle Star Indicates Parallel Evolution Influenced by Genomic Availability of Haloalkane Dehalogenase," *Molecular Biology and Evolution*, vol. 42, no. 5, Art. no. msaf081, 2025. [Online]. Available: https://doi.org/10.1093/molbev/msaf081